ULTRON — Project Details

Personal AI assistant for Windows and Android · Status as of 10 November 2026 · Repository: github.com/nityamsharma19022-ux/ultron-ai-assistant (branch claude/fervent-lovelace-fqank5)

1. What ULTRON is

ULTRON is a personal AI assistant and "AI operating system" for one owner. A central server (ULTRON Core) runs on your Windows PC in Docker. The ULTRON app (Windows and Android) talks only to Core. A Windows agent carries out actions on the PC, and the Android app does the same on the phone. Core chooses an AI model for every message (cloud first, local fallback), remembers what you allow, plans and runs tasks with permissions, and sends proactive notifications.

2. Programming languages and technologies

Part	Language	Main technologies
ULTRON Core (server)	Python 3.13 (needs 3.12+)	FastAPI, Uvicorn, SQLAlchemy 2.1 (async), Alembic, asyncpg, Pydantic 2, httpx, structlog, PyJWT, Argon2 (pwdlib), cryptography, pypdf, python-docx, openpyxl, Anthropic SDK
Database / cache	SQL (PostgreSQL)	PostgreSQL (full-text search), Redis (rate limits, pairing codes, live device metrics)
ULTRON app (Windows + Android)	Dart (Flutter 3.47.9)	Riverpod 3, http, flutter_secure_storage, shared_preferences, speech_to_text, flutter_tts, file_picker, crypto
Android system parts	Kotlin	VoiceInteractionService (digital assistant), recognition service, phone tools (apps, alarms, timers, notifications, status)
Windows agent	Python 3.12+	httpx, psutil, send2trash, pypdf, python-docx; Windows APIs via PowerShell for toasts
Setup / operations	PowerShell, YAML	Docker Compose (postgres, redis, core, optional ollama), Windows helper scripts, GitHub Actions CI

3. AI models and providers

All AI access goes through ULTRON Core; API keys stay in Core's .env file and are never sent to the apps. Every provider is optional; the ones without a key are skipped automatically.

Order	Provider	Default model	Type	Notes
1	FreeLLMAPI (your gateway, 127.0.0.1:31415)	auto (Instant: auto:fast, Deep: auto:smart)	Cloud	Router on your PC that forwards to free tiers of many providers (your list includes Gemini 3.x Flash, Qwen 3.8 27B, gpt-oss, Gemma). Records the real model used.
2	Groq	llama-3.3-70b-versatile	Cloud	Free tier
3	Google Gemini	gemini-2.5-flash	Cloud	Free tier
4	OpenRouter	(you choose)	Cloud	Free ":free" models
5	Custom	(you choose)	Cloud	Any OpenAI-compatible API
6	Anthropic Claude	claude-opus-5-5	Cloud	Paid; only if you add a key
Last	Ollama (local, on your PC)	qwen3.5:4b	Local	Works offline; the only AI used for private conversations

4. How ULTRON chooses the AI

- Cloud first, local fallback: providers are tried in your order; on rate limits or outages a provider is paused briefly and the next one is tried; offline, Ollama answers.

- Private conversations use local AI only, never a cloud provider (FreeLLMAPI counts as cloud).
- By kind of request: chat, coding, reasoning, writing, summarizing, planning; preferred providers per kind can be set (ULTRON_AI_TASK_PROFILES).
- Response modes: Instant (fast model, short answers), Balanced (default), Deep reasoning (strongest routing, longer answers).
- Operating modes: Normal, Work, Developer, Study, Focus and custom ones change tone, preferred AI, memory use and what ULTRON may act on.
- Speed is measured: time to first word and total time for every reply. ULTRON Core itself adds only about 0.02 s; the rest is the AI model.
- Safety: models only return text. Actions happen through ULTRON's own validated planner and permission system.

5. Main features

Area	Features
Command Center (Home)	12-hour clock and greeting ("Good evening, Sir"), command bar, quick commands, live device panel (CPU, RAM, disk, battery, network), tasks, reminders, notifications, AI activity, autonomy sliders; floating draggable panels on desktop
Chat	Streaming replies, CLOUD/LOCAL badge, stop/retry, private chats, conversation search, documents as context, reply timing
Memory	Long-term memories (add, edit, pin, private, project tags, pause saving, export, delete), recalled into chat as background facts
Tasks and automation	Commands become plans; permission classes; autonomy levels 0–5 per area (Core, PC, phone); approvals; retries; schedules; reminders; emergency stop; automatic safety pause
Windows agent	Open, switch, close and list apps; files (list, search, read, write, copy, rename, move, organize, Recycle Bin, undo) in allowed folders only; workflows; system status; live metrics; Windows notifications; open web links
Android	Digital assistant (system gesture), voice, phone tools (open apps, alarms, timers, notifications, status), battery/network metrics
Proactive	Notifications with priorities, quiet hours, daily briefing; alerts for failed tasks, waiting approvals, reminders, offline devices, low battery
Voice	Push-to-talk, read-aloud, interruptible speech; privacy rules for private chats
Documents and web	PDF, Word, Excel and text understanding; safe web page reading (private network addresses refused)
Personality and time	Humble, warm, polite; addresses you as "Sir" (changeable); 12-hour time everywhere by default, optional 24-hour
Security	Owner account (Argon2), device tokens with rotation, pairing codes, HTTPS with pinned certificate, audit log, data export

6. Data and storage

- PostgreSQL holds users, devices, conversations, messages, memories, tasks, steps, permissions, reminders, schedules, documents, modes, notifications and settings (9 database migrations).
- Redis holds short-lived data: rate limits, pairing codes and live device metrics (3 minutes).
- Your data lives in Docker volumes on your PC. Backups: pg_dump, or Settings → Your data → Export all data (JSON, no passwords or tokens).

7. Size, tests and quality checks

Item	Value
Code (excluding tests)	Core ≈ 11,300 lines Python · App ≈ 10,900 lines Dart · Agent ≈ 1,550 lines Python · Android ≈ 370 lines Kotlin

Item	Value
Automated tests	Server 236 · App 124 · Agent 36 (+1 Windows-only) — all passing on the last run
Continuous integration	GitHub Actions: Core CI (Linux), App CI (tests, Android release APK, Windows build), Agent CI (Linux + Windows) — all green
Git history	30 commits on the development branch; main is untouched

8. What is verified on your PC

- ULTRON Core running in Docker: ready, database current, Redis OK, local AI (qwen3.5:4b) OK.
- FreeLLMAPI gateway answering with working models.
- Windows app: sign-in, Home screen, Core online, CLOUD and LOCAL indicators.
- Windows agent connected (19 tools) with live CPU, memory, battery and disk shown on Home.

9. Not built yet

- Windows tray icon, background service, wake word and keyboard shortcut to talk.
- Meaning-based (semantic) memory search; image understanding; offline sync between devices.
- Keyboard and mouse automation; screenshot understanding.
- Weather and calendar widgets; Google Calendar, Gmail, Drive, OneDrive and GitHub connections (each needs your sign-in).
- Android phone test (APK builds, not yet installed on the iQOO).

Full details are in the repository's docs folder: ULTRON_ARCHITECTURE, FEATURE_STATUS, ULTRON_ROADMAP, AI_BRAIN_REPORT, PERFORMANCE_REPORT, VOICE_SYSTEM_REPORT, KNOWN_ISSUES and CHANGELOG.